

Efficient Estimation of Generalized Partially Linear Single-Index Models Using Penalized Splines

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1 Introduction

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2.1 Model

In this manuscript, we investigate penalized spline estimation for generalized partially linear single-index models (GPLSIMs). Given covariate vectors $\mathbf{X} = \mathbf{x}$ and $\mathbf{Z} = \mathbf{z}$, we assume that the conditional density of the response variable Y belongs to an exponential family

$$f(y|\mathbf{x}, \mathbf{z}) = \exp \left\{ \frac{y\theta - b(\theta)}{\phi} + c(y, \phi) \right\}, \quad (1)$$

for known functions $b(\cdot)$ and $c(\cdot)$, where $\theta = \theta(\mathbf{x}, \mathbf{z})$ denotes the canonical parameter and ϕ is a scale parameter in generalized linear models (GLMs). Under this specification, the conditional mean is $\mu = b'(\theta)$ and the conditional variance satisfies $\text{var}(Y|\mathbf{x}, \mathbf{z}) = \phi b''(\theta) = \phi V(\mu)$, where $V(\cdot)$ is the variance function (see McCullagh and Nelder 1989 for further details). The GPLSIM is formulated as

$$g\{\mu(\mathbf{x}, \mathbf{z})\} = \boldsymbol{\beta}^\top \mathbf{x} + \varphi(\boldsymbol{\alpha}^\top \mathbf{z}), \quad (2)$$

where $\boldsymbol{\alpha} \in \mathbb{R}^p$ and $\boldsymbol{\beta} \in \mathbb{R}^d$ denote the index parameter vector and the regression coefficient vector, respectively. The function $\varphi(\cdot)$ is an unknown smooth function, and $g(\cdot)$ is a known link function, as in GLMs. When $g = (b')^{-1}$, the link is canonical.

It is well known that the index parameter vector $\boldsymbol{\alpha}$ in model (2) is not identifiable. A common remedy is to impose a normalization constraint. Specifically, we standardize $\boldsymbol{\alpha}$ such that $\|\boldsymbol{\alpha}\| = 1$ and adopt a re-parameterization that removes this constraint by expressing

$$\boldsymbol{\alpha}_\eta = \frac{(1, \eta_1, \dots, \eta_{p-1})^\top}{\sqrt{1 + \|\boldsymbol{\eta}\|^2}}, \quad \boldsymbol{\eta} = (\eta_1, \dots, \eta_{p-1})^\top,$$

where $\|\cdot\|$ denotes the Euclidean norm. This re-parameterization enables unconstrained optimization; see the Appendix of Yu et al. (2017) for additional details.

Let $(y_1, \mathbf{x}_1, \mathbf{z}_1), \dots, (y_n, \mathbf{x}_n, \mathbf{z}_n)$ be a random sample. Under the GPLSIM (2), the log-likelihood for the unknown parameter **vector** $\boldsymbol{\tau} = (\boldsymbol{\beta}^\top, \boldsymbol{\eta}^\top, \varphi)^\top$ is

$$l(\boldsymbol{\tau}) = \sum_{i=1}^n \left\{ \frac{y_i \theta_i - b(\theta_i)}{\phi} + c(y_i, \phi) \right\},$$

where $\theta_i = \theta(\mathbf{x}_i, \mathbf{z}_i)$. Under the canonical link, $\theta_i = \boldsymbol{\beta}^\top \mathbf{x}_i + \varphi(\boldsymbol{\alpha}_\eta^\top \mathbf{z}_i)$. Penalized methods are widely used in semiparametric regression to regulate the smoothness of unknown functions and to mitigate overfitting; see, for example, Ma and Kosorok (2010), Lu and Li (2017), and Lu et al. (2022), among many others. Motivated by this, we estimate $\boldsymbol{\tau}$ by maximizing the penalized log-likelihood

$$l_\lambda(\boldsymbol{\tau}) = l(\boldsymbol{\tau}) - \frac{1}{2\phi} \lambda J^2(\varphi), \quad (3)$$

where $\lambda > 0$ is a (possibly data-dependent) smoothing parameter that governs the trade-off between model fidelity and smoothness of the fitted curve, and $J^2(\varphi) = \int \{\varphi^{(m)}(u)\}^2 du$ is the roughness penalty for a fixed integer $m \geq 1$.

Direct maximization of the penalized log-likelihood (3) is typically difficult, as it involves simultaneous estimation of regression parameters and an infinite-dimensional function. Following Lu and Loomis (2013), we approximate the **unknown** function $\varphi(\cdot)$ using a constrained B -spline representation when $\varphi(\cdot)$ is sufficiently smooth. Specifically,

$$\varphi(\cdot) \approx \varphi_n(\cdot) = \sum_{j=1}^{q_n} \gamma_j s_j(\cdot) = \boldsymbol{\gamma}^\top \mathbf{s}(\cdot),$$

where $\mathbf{s}(\cdot) = (s_1(\cdot), \dots, s_{q_n}(\cdot))^\top$ is the vector of B -spline basis functions and $\boldsymbol{\gamma} = (\gamma_1, \dots, \gamma_{q_n})^\top$ denotes the associated spline coefficients.

Clearly, the spline function $\varphi_n(\cdot)$ is identifiable only up to an additive constant. To ensure identifiability, we impose a sum-to-zero constraint on $\varphi_n(\cdot)$, namely $\sum_{i=1}^n \varphi_n(u_i) = \mathbf{S}^\top \boldsymbol{\gamma} = 0$ for $u_i = \boldsymbol{\alpha}_\eta^\top \mathbf{z}_i$, where $\mathbf{S} = \sum_{i=1}^n \mathbf{s}_i$ and $\mathbf{s}_i = \mathbf{s}(u_i) = (s_1(u_i), \dots, s_{q_n}(u_i))^\top$. To facilitate implementation, we re-parameterize the q_n constrained spline coefficients $\boldsymbol{\gamma}$ in terms of $(q_n - 1)$ unconstrained parameters $\boldsymbol{\zeta} = (\zeta_1, \dots, \zeta_{q_n-1})^\top$. We achieve this through a QR-decomposition of $\mathbf{S}$. **Specifically**, let $\mathbf{S} = \mathbf{Q}\mathbf{R}$, where $\mathbf{Q}$ is a $q_n \times q_n$ orthogonal matrix and $\mathbf{R}$ is a $q_n \times 1$ upper triangular matrix. Partition $\mathbf{Q} = (\mathbf{Q}_1, \mathbf{Q}_2)$, where $\mathbf{Q}_2$ is a $q_n \times (q_n - 1)$ matrix. Since $\mathbf{S}^\top (\mathbf{Q}_2 \boldsymbol{\zeta}) = 0$ for any $\boldsymbol{\zeta} \in \mathbb{R}^{q_n-1}$, choosing $\boldsymbol{\gamma} = \mathbf{Q}_2 \boldsymbol{\zeta}$ automatically enforces the sum-to-zero constraint. It then follows that $\varphi_n(u_i) = \boldsymbol{\gamma}^\top \mathbf{s}_i = \boldsymbol{\zeta}^\top \mathbf{b}_i$ for $\mathbf{b}_i = \mathbf{Q}_2^\top \mathbf{s}_i$. For further discussion of re-parameterizing parameters subject to linear constraints, see Wood (2017).

Under the above specifications, we obtain the spline model

$$g\{\mu(\mathbf{x}_i, \mathbf{z}_i)\} = \boldsymbol{\beta}^\top \mathbf{x}_i + \boldsymbol{\zeta}^\top \mathbf{b}_i, \quad (4)$$

for $\mathbf{b}_i = \mathbf{Q}_2^\top \mathbf{s}(\boldsymbol{\alpha}_\eta^\top \mathbf{z}_i)$. The full set of unknown parameters to be estimated is therefore $\boldsymbol{\theta} = (\boldsymbol{\beta}^\top, \boldsymbol{\eta}^\top, \boldsymbol{\zeta}^\top)^\top$.

Under the spline approximation, where $\varphi(\cdot)$ is replaced by $\varphi_n(\cdot)$, the penalty matrix $J^2(\varphi_n)$ becomes a band matrix whose (k, l) th element is $\int b_k^{(m)}(u)b_l^{(m)}(u)du$. To facilitate computation, we adopt the P -spline approach of Eilers and Marx (1996). In particular, the penalty matrix is approximated by $\mathbf{D}^\top \mathbf{D}$, where $\mathbf{D}$ is the difference matrix generated by the m th-order difference operator Δ . In both the simulation study and the real data analysis, we set $m = 2$. Under this specification, the difference penalty term $\boldsymbol{\zeta}^\top \mathbf{D}^\top \mathbf{D} \boldsymbol{\zeta}$ simplifies to $\sum_{k=3}^{q_n-1} (\Delta^2 \zeta_k)^2$, where $\Delta^2 \zeta_k = \zeta_k - 2\zeta_{k-1} + \zeta_{k-2}$. The P -spline formulation provides considerable flexibility for penalized estimation and is easily implemented using standard software; see Wood (2017, p. 206) for additional details.

Proceeding in this manner, we obtain the penalized spline log-likelihood for $\boldsymbol{\theta}$,

$$l_\lambda(\boldsymbol{\theta}) = \sum_{i=1}^n \left\{ \frac{y_i \theta_i - b(\theta_i)}{\phi} + c(y_i; \theta_i) \right\} - \frac{\lambda}{2\phi} \boldsymbol{\zeta}^\top \mathbf{D}^\top \mathbf{D} \boldsymbol{\zeta}. \quad (5)$$

The penalized spline estimator $\hat{\boldsymbol{\theta}} = (\hat{\boldsymbol{\beta}}^\top, \hat{\boldsymbol{\eta}}^\top, \hat{\boldsymbol{\zeta}}^\top)^\top$ is defined as the maximizer of (5), and the resulting estimator of $\psi(\cdot)$ is $\hat{\varphi}(\cdot) = \varphi_n(\cdot | \hat{\boldsymbol{\zeta}}, \hat{\boldsymbol{\eta}})$. This penalized spline framework is straightforward to implement, as standard software can be used for model fitting. Moreover, the regression parameters $(\boldsymbol{\beta}, \boldsymbol{\eta})$ and the spline coefficients $\boldsymbol{\zeta}$ are estimated jointly, making the procedure computationally efficient.

2.2 Algorithm

Let $\mathbf{b}'_i = \mathbf{Q}_2^\top(s'_1(u_i), \dots, s'_{q_n}(u_i))^\top$ for $u_i = \boldsymbol{\alpha}_i^\top \mathbf{z}_i$, where $s'_j(\cdot)$ denotes the first derivative of $s_j(\cdot)$ for $1 \leq j \leq q_n$. Define $\boldsymbol{\delta}_i = (\delta_{i1}, \dots, \delta_{i(p-1)})^\top$, where

$$\delta_{il} = (1 + \|\boldsymbol{\eta}\|^2)^{-1/2} z_{i(l+1)} - (1 + \|\boldsymbol{\eta}\|^2)^{-3/2} \eta_l \left(z_{i1} + \sum_{j=1}^{p-1} z_{i(j+1)} \eta_j \right),$$

for $l = 1, \dots, p-1$. Let $\boldsymbol{\xi}_i = (\xi_{i1}, \dots, \xi_{i(p-1)})^\top$ with $\xi_{il} = \delta_{il} \boldsymbol{\zeta}^\top \mathbf{b}'_i$. Set $\mathbf{P} = \text{diag}\{\mathbf{0}_{(d+p-1) \times (d+p-1)}, \mathbf{D}^\top \mathbf{D}\}$ and $\mathbf{E} = (\mathbf{E}_1, \dots, \mathbf{E}_n)^\top$, where $\mathbf{E}_i = (\mathbf{x}_i^\top, \boldsymbol{\xi}_i^\top, \mathbf{b}_i^\top)^\top$. Also denote $\mathbf{y} = (y_1, \dots, y_n)^\top$ and $\boldsymbol{\mu} = (\mu_1, \dots, \mu_n)^\top$.

To maximize the log-likelihood via the Fisher scoring method, we require the gradient vector and Fisher information matrix of $l_\lambda(\boldsymbol{\theta})$. Define $\mathbf{G}_i = \partial \mu_i / \partial \boldsymbol{\theta}$. Clearly, $\mathbf{G}_i = g'^{-1}(\mu_i) \mathbf{E}_i$. Let $\mathbf{G} = (\mathbf{G}_1, \dots, \mathbf{G}_n)^\top$ and $\mathbf{V} = \text{diag}\{V(\mu_1), \dots, V(\mu_n)\}$.

Under the above specifications, the score function and Fisher information matrix are

$$\nabla l_\lambda(\boldsymbol{\theta}) = \phi^{-1} \mathbf{G}^\top \mathbf{V}^{-1} (\mathbf{y} - \boldsymbol{\mu}) - \phi^{-1} \lambda \mathbf{P} \boldsymbol{\theta}$$

and

$$\mathbf{I}_\lambda(\boldsymbol{\theta}) = \phi^{-1} \mathbf{G}^\top \mathbf{V}^{-1} \mathbf{G} + \phi^{-1} \lambda \mathbf{P}.$$

Clearly $\mathbf{I}_\lambda(\boldsymbol{\theta})$ is positive definite. As discussed later in the proposed two-stage iterative algorithm, this property ensures that the updated smoothing parameter remains positive.

To carry out the penalized estimation, we propose a two-stage iterative algorithm for obtaining the penalized spline estimator $\hat{\boldsymbol{\theta}}$. In the inner iteration, the current estimate $\boldsymbol{\theta}_\lambda^{(l)}$ is updated to $\boldsymbol{\theta}_\lambda^{(l+1)}$ using

$$\mathbf{G}^\top = \mathbf{E}^\top \text{diag}\{g'^{-1}(\mu_i)\}$$

the Fisher scoring method with a step-halving line search. Specifically,

$$\boldsymbol{\theta}_\lambda^{(l+1)} = \boldsymbol{\theta}_\lambda^{(l)} + (1/2)^k \Delta_\lambda^{(l)} \quad (6)$$

for $\Delta_\lambda^{(l)} = \mathbf{I}_\lambda^{-1}(\boldsymbol{\theta}_\lambda^{(l)}) \nabla l_\lambda(\boldsymbol{\theta}_\lambda^{(l)})$, where $\mathbf{I}_\lambda(\boldsymbol{\theta}_\lambda^{(l)})$ and $\nabla l_\lambda(\boldsymbol{\theta}_\lambda^{(l)})$ denote the Fisher information matrix and the score function evaluated at $\boldsymbol{\theta}_\lambda^{(l)}$, respectively. The integer k is the smallest non-negative value satisfying

$$l_\lambda(\boldsymbol{\theta}_\lambda^{(l)} + (1/2)^k \Delta_\lambda^{(l)}, \phi^{(l)}) > l_\lambda(\boldsymbol{\theta}_\lambda^{(l)}, \phi^{(l)}).$$

Further, update ϕ using an adjusted Pearson statistic, namely,

$$\phi^{(l+1)} = \frac{1}{\rho} \sum_{i=1}^n \frac{(y_i - \hat{\mu}_i)^2}{V(\hat{\mu}_i)}, \quad (7)$$

where ρ denotes the effective degrees of freedom in penalized estimation and is commonly used to correct bias in estimating ϕ . An approximation for the effective degrees of freedom is $\rho = \text{tr}\{\mathbf{I}_\lambda(\boldsymbol{\theta})^{-1} \mathbf{I}(\boldsymbol{\theta})\}$, where $\mathbf{I}(\boldsymbol{\theta}) = \mathbf{I}_\lambda(\boldsymbol{\theta})|_{\lambda=0}$; see Wood et al. (2016, pp. 1550) for further discussion of effective degrees of freedom in a general setting.

During the outer iteration, the smoothing parameter λ is updated using the generalized Fellner–Schall method (Wood and Fasiolo, 2017), namely,

$$\bar{\lambda} = \frac{\text{tr}\{\mathbf{P}_\lambda^- \mathbf{P}\} - \text{tr}\{\mathbf{I}_\lambda^{-1}(\boldsymbol{\theta}_\lambda^{(l+1)}) \mathbf{P}\}}{\boldsymbol{\theta}_\lambda^{(l+1)\top} \mathbf{P} \boldsymbol{\theta}_\lambda^{(l+1)}} \lambda, \quad (8)$$

where $\mathbf{P}_\lambda^-$ is the generalized inverse of $\mathbf{P}_\lambda = \lambda \mathbf{P}$. Since $\mathbf{I}_\lambda(\boldsymbol{\theta}_\lambda^{(l+1)})$ is positive definite, Theorem 1 of Wood and Fasiolo (2017) ensures that the update of λ in (8) is positive. Once λ is updated, the inner iteration restarts to obtain updated estimates of $\boldsymbol{\theta}$ and ϕ for the new value $\bar{\lambda}$. The two-stage algorithm proceeds in this manner until convergence. The outline of the algorithm is summarized as follows:

Step 0. (a) Let $\boldsymbol{\alpha}^{(0)}$ be the estimator of the linear model $g\{E(Y_i|\mathbf{z}_i, \mathbf{x}_i)\} = \boldsymbol{\beta}^\top \mathbf{x}_i + \boldsymbol{\alpha}^\top \mathbf{z}_i$. Standardize $\boldsymbol{\alpha}^{(0)}$ as the initial estimator $\boldsymbol{\alpha}_\eta^{(0)}$ for $\boldsymbol{\alpha}_\eta$; (b) given the preliminary estimators of index values $\{u_i^{(0)} = \mathbf{z}_i^\top \boldsymbol{\alpha}_\eta^{(0)} : i = 1, \dots, n\}$, fit the generalized partially linear model for $(\boldsymbol{\beta}, \boldsymbol{\zeta})$

$$g\{E(Y_i|\mathbf{z}_i, \mathbf{x}_i)\} = \boldsymbol{\beta}^\top \mathbf{x}_i + \boldsymbol{\zeta}^\top \mathbf{b}_i^{(0)}$$

to get $(\boldsymbol{\beta}^{(0)}, \boldsymbol{\zeta}^{(0)})$, where $\mathbf{b}_i^{(0)} = \{b_1(u_i^{(0)}), \dots, b_{q_n-1}(u_i^{(0)})\}^\top$; (c) calculate the initial value of ϕ via (7) and set the initial value of λ to be 1.

Step 1 (Inner iteration). For a given $\lambda > 0$, update the current $\boldsymbol{\theta}_\lambda^{(l)}$ to $\boldsymbol{\theta}_\lambda^{(l+1)}$ through the Fisher's scoring approach (6) and calculate ϕ via equation (7) at $\boldsymbol{\theta} = \boldsymbol{\theta}_\lambda^{(l+1)}$.

Step 2 (Outer iteration). Update λ to $\bar{\lambda}$ through the generalized Fellner–Schall method via (8) and back to **Step 1** to identify $\boldsymbol{\theta}_{\bar{\lambda}}^{(l+1)}$. Repeat the iteration until $\|\boldsymbol{\theta}_{\bar{\lambda}}^{(l+1)} - \boldsymbol{\theta}_{\bar{\lambda}}^{(l+1)}\| < 10^{-6}$.

As the algorithm converges, set $\hat{\boldsymbol{\theta}} = \boldsymbol{\theta}_{\bar{\lambda}}^{(l+1)}$ and $\hat{\phi} = \phi_{\bar{\lambda}}^{(l+1)}$ as the penalized spline estimator of $\boldsymbol{\theta}$ and ϕ and $\hat{\varphi}(\cdot) = \varphi_n(\cdot|\hat{\boldsymbol{\zeta}}, \hat{\boldsymbol{\eta}})$ is the penalized spline estimator of $\varphi(\cdot)$ accordingly.

Step 0. (a) Let $\alpha^{(0)}$ be the estimator from the linear model

$$g\{E(Y_i|\mathbf{z}_i, \mathbf{x}_i)\} = \beta^\top \mathbf{x}_i + \alpha^\top \mathbf{z}_i.$$

Standardize $\alpha^{(0)}$ to obtain the initial estimator $\alpha_\eta^{(0)}$ for α_η . (b) Given the preliminary index values $\{u_i^{(0)} = \mathbf{z}_i^\top \alpha_\eta^{(0)} : i = 1, \dots, n\}$, fit the generalized partially linear model for (β, ζ)

$$g\{E(Y_i|\mathbf{z}_i, \mathbf{x}_i)\} = \beta^\top \mathbf{x}_i + \zeta^\top \mathbf{b}_i^{(0)}$$

to obtain $(\beta^{(0)}, \zeta^{(0)})$, where $\mathbf{b}_i^{(0)} = \{b_1(u_i^{(0)}), \dots, b_{q_n-1}(u_i^{(0)})\}^\top$. (c) Compute the initial value of ϕ using (7) and set the initial value of λ to 1.

Step 1. (Inner iteration) For a given $\lambda > 0$, update the current estimate $\theta_\lambda^{(l)}$ to $\theta_\lambda^{(l+1)}$ using the Fisher scoring update (6), and evaluate ϕ via (7) at $\theta = \theta_\lambda^{(l+1)}$.

Step 2. (Outer iteration) Update λ to $\bar{\lambda}$ using the generalized Fellner–Schall method (8), then return to **Step 1** to compute $\theta_{\bar{\lambda}}^{(l+1)}$. Repeat the iteration until

$$\|\theta_{\bar{\lambda}}^{(l+1)} - \theta_{\bar{\lambda}}^{(l)}\| < 10^{-6}.$$

As the algorithm converges, set $\hat{\theta} = \theta_{\bar{\lambda}}^{(l+1)}$ and $\hat{\phi} = \phi_{\bar{\lambda}}^{(l+1)}$ as the penalized spline estimators of θ and ϕ , respectively, and let $\hat{\varphi}(\cdot) = \varphi_n(\cdot|\hat{\zeta}, \hat{\eta})$ be the corresponding penalized spline estimator of $\varphi(\cdot)$.

For practical computation, we implement the Fisher scoring update (6) using weighted penalized least squares regression. In particular, define $\mathbf{X} = \mathbf{V}^{-1/2}\mathbf{G}$, where $\mathbf{V}^{-1/2}$ is a square root of $\mathbf{V}^{-1}$, that is, $\mathbf{V}^{-1/2\top}\mathbf{V}^{-1/2} = \mathbf{V}^{-1}$. For given λ and ϕ , a single Fisher scoring update takes the form

$$\theta_\lambda^{(l+1)} = (\mathbf{X}^\top \mathbf{X} + \mathbf{P}_\lambda)^{-1} \mathbf{X}^\top \mathbf{z}$$

for $\mathbf{z} = \mathbf{X}\theta_\lambda^{(l)} + \mathbf{V}^{-1/2}(\mathbf{y} - \boldsymbol{\mu})$. Hence, given λ and ϕ , the update $\theta_\lambda^{(l+1)}$ minimizes the penalized least squares function

$$\|\mathbf{z} - \mathbf{X}\theta\|^2 + \theta^\top \mathbf{P}_\lambda \theta, \tag{9}$$

which can also be expressed as $\|\tilde{\mathbf{z}} - \tilde{\mathbf{X}}\theta\|^2$, where $\tilde{\mathbf{z}} = (\mathbf{z}^\top, \mathbf{0}^\top)^\top$ and $\tilde{\mathbf{X}} = (\mathbf{X}^\top, \mathbf{P}_\lambda^{1/2\top})^\top$ with $\mathbf{P}_\lambda^{1/2\top}\mathbf{P}_\lambda^{1/2} = \mathbf{P}_\lambda$. In other words, the Fisher scoring update reduces to a standard linear regression problem.

To enhance numerical stability, an orthogonal matrix method based on pivoted QR decomposition of $\tilde{\mathbf{X}}$ can be used to solve the linear model. Specifically, write $\tilde{\mathbf{X}} = \mathbf{Q} \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix} = \mathbf{Q}_f \mathbf{R}$, where $\mathbf{Q}$ is an orthogonal matrix and $\mathbf{R}$ is a triangular matrix, and $\mathbf{Q}_f$ consists of the first $(d + q_n + p - 2)$ columns of $\mathbf{Q}$. It then follows that

$$\hat{\theta} = (\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} \tilde{\mathbf{X}}^\top \tilde{\mathbf{z}} = \mathbf{R}^{-1} \mathbf{Q}_f^\top \tilde{\mathbf{z}}$$

is the least squares estimator of θ ; see Wood (2016, p. 168) for additional details.

2.3 Selection of the knots

2.4 Interpretation of the model

3 Asymptotic results

We will establish

- Uniform consistency and rate of convergence of $\hat{\varphi}(\cdot)$
- Asymptotic normality of $(\hat{\beta}, \hat{\eta})$

4 Variance Estimation

4.1 Consistent Variance Estimator

4.2 Robust Variance Estimator

Accurate variance estimation for the regression parameters (β, η) is essential for valid statistical inference. Under the penalized likelihood framework, the asymptotic covariance matrix of the estimator $\hat{\theta} = (\hat{\beta}^\top, \hat{\eta}^\top, \hat{\zeta}^\top)^\top$ is typically obtained from the inverse of the penalized Fisher information matrix

$$\mathbf{V}_\lambda(\hat{\theta}) = \mathbf{I}_\lambda^{-1}(\hat{\theta}) = \phi(\mathbf{G}^\top \mathbf{V}^{-1} \mathbf{G} + \lambda \mathbf{P})^{-1}.$$

The block of $\mathbf{V}_\lambda(\hat{\theta})$ corresponding to (β, η) provides a natural estimator of the covariance matrix of $(\hat{\beta}, \hat{\eta})$ under the assumed working model.

However, it is well recognized that penalized spline components introduce additional smoothing bias, which leads to systematic underestimation of the variability of the index parameter estimator $\hat{\eta}$ when relying solely on the inverse Fisher information. This issue is particularly notable in partially linear single-index logistic models, where the spline approximation of the unknown link function interacts nontrivially with the index structure, thereby inflating the true sampling variability of $\hat{\eta}$ beyond what is captured by the naive Fisher information-based approach.

To address this underestimation, we propose a robust sandwich-type variance estimator that accommodates potential model misspecification and accounts for the uncertainty induced by spline smoothing. In general, a robust covariance estimator can be expressed as

$$\mathbf{V}_{\lambda,r}(\hat{\theta}) = \mathbf{I}_\lambda^{-1}(\theta) \mathbf{H}(\theta) \mathbf{I}_\lambda^{-1}(\theta),$$

where

$$\mathbf{H}(\theta) = \text{var}(\nabla l_\lambda(\theta)) = \phi^{-2} \sum_{i=1}^n \nu_i^{-2} \text{var}(y_i) \mathbf{G}_i^{\otimes 2},$$

for $\nu_i = V(\mu_i)$, denotes the score covariance matrix formed from the individual score contributions. The variance $\text{var}(y_i)$ is typically estimated by the moment estimator $\hat{r}_i^2 = (y_i - \hat{\mu}_i)^2$. The corresponding variance estimator for $(\hat{\beta}, \hat{\eta})$ is obtained by extracting the appropriate submatrix of the expression above. This robust adjustment substantially improves the accuracy of variance estimation for (β, η) , especially for the index parameter η , whose variance is typically underestimated under the naive Fisher information approach.

4.3 Bias-Corrected Robust Variance Estimator

Although the sandwich variance estimator yields asymptotically valid inference for generalized single-index models, it is well documented that this estimator can suffer from substantial downward bias in finite samples, particularly for parameters that enter the model nonlinearly. This phenomenon is especially evident for the index parameter in logistic single-index models, where estimation uncertainty is amplified by the nonlinear mean structure and the normalization constraint imposed on the index. Consequently, standard sandwich-based variance estimates often fail to adequately capture the true sampling variability of the index parameter, leading to overly optimistic inference. To address this issue, we adopt finite-sample variance correction methods proposed by Mancl and DeRouen (2001) and by Morel et al. (2003). These corrections explicitly account for the impact of parameter estimation on the estimating equations and adjust the sandwich estimator to reduce finite-sample bias. Incorporating these variance correction techniques within the generalized single-index modeling framework provides a principled and computationally tractable approach to improving inference for index parameters, particularly in logistic models where curvature and nonlinearity exacerbate variance underestimation.

4.3.1 Leverage-Adjusted Robust Estimator

To calculate the robust covariance estimator, the residuals, $\hat{r}_i = y_i - \hat{\mu}_i$, are used to estimate $\text{var}(y_i)$, i.e., $\text{var}(y_i) \approx \hat{r}_i^2$. It is well known that the residuals tend to be too small. Hence, the robust estimator would be expected to underestimate the covariance of $\hat{\theta}$, and the bias of the robust estimator due to underestimation of $\text{var}(y_i)$ would be large when the bias of the residuals is large.

An alternative robust covariance estimator for $\hat{\theta}$ is proposed that is intended to reduce the bias of the residual estimator, $\hat{r}_i^2$. To derive an approximation for the bias of $\hat{r}_i^2$, consider a first-order Taylor series expansion of the residual r_i about θ given by

$$\hat{r}_i = r_i - \mathbf{G}_i^\top (\hat{\theta} - \theta), \quad (10)$$

where $r_i = r_i(\theta) = y_i - \mu_i$. By squaring (10) and taking the expectation, one obtains

$$E(\hat{r}_i^2) = E(r_i^2) + E\left\{\mathbf{G}_i^\top (\hat{\theta} - \theta)^{\otimes 2} \mathbf{G}_i\right\} - 2E\left\{r_i \mathbf{G}_i^\top (\hat{\theta} - \theta)\right\}.$$

Using the first-order approximation,

$$\nabla l_\lambda(\hat{\theta}) = \nabla l_\lambda(\theta) - \phi^{-1} \sum_{i=1}^n \nu_i^{-1} \mathbf{G}_i^{\otimes 2} (\hat{\theta} - \theta),$$

$$\mathbf{G}(\hat{\theta}) - \mathbf{G}(\theta) = o_p(1)$$

and hence,

$$\hat{\boldsymbol{\theta}} - \boldsymbol{\theta} = \left(\sum_{i=1}^n \nu_i^{-1} \mathbf{G}_i^{\otimes 2} \right)^{-1} \sum_{i=1}^n \nu_i^{-1} r_i \mathbf{G}_i. \quad (11)$$

We can substitute (11) into (10) to obtain

$$\hat{r}_i = r_i - \sum_{j=1}^n \nu_j^{-1} \left(\mathbf{G}_i^T \mathbf{A} \mathbf{G}_j \right) r_j = r_i - \sum_{j=1}^n h_{ij} r_j,$$

where $\mathbf{A} = \left(\sum_{i=1}^n \nu_i^{-1} \mathbf{G}_i^{\otimes 2} \right)^{-1}$. Separating the $j = i$ term yields

$$\hat{r}_i = (1 - h_{ii}) r_i - \sum_{j \neq i} h_{ij} r_j,$$

and hence,

$$\hat{r}_i^2 = (1 - h_{ii})^2 r_i^2 - 2(1 - h_{ii}) r_i \left(\sum_{j \neq i} h_{ij} r_j \right) + \left(\sum_{j \neq i} h_{ij} r_j \right)^2.$$

Assume that $r_1, \dots, r_n$ are independent across subjects, so that $E(r_i r_j) = 0$, for $j \neq i$. Taking expectation of the above equation, the cross term vanishes, and we get

$$E(\hat{r}_i^2) = (1 - h_{ii})^2 E(r_i^2) + \sum_{j \neq i} h_{ij}^2 E(r_j^2)$$

Because $E(r_i^2) = \text{var}(y_i)$, this becomes

$$E(\hat{r}_i^2) = (1 - h_{ii})^2 \text{var}(y_i) + \sum_{j \neq i} h_{ij}^2 \text{var}(y_j).$$

Noting that the elements of h_{ij} are between 0 and 1 and typically close to 0, so that we drop the last summation and obtain

More justification later

$$E(\hat{r}_i^2) \approx (1 - h_{ii})^2 \text{var}(y_i), \quad (12)$$

which is the main leverage-adjusted form used in the alternative robust covariance estimator.

The robust covariance estimator takes the form

$$\text{var}(\hat{\boldsymbol{\theta}}) = \phi^{-2} \mathbf{I}_{\lambda}^{-1}(\boldsymbol{\theta}) \left\{ \sum_{i=1}^n \nu_i^{-2} \text{var}(y_i) \mathbf{G}_i^{\otimes 2} \right\} \mathbf{I}_{\lambda}^{-1}(\boldsymbol{\theta}).$$

However, the true covariance $\text{var}(y_i)$ is unknown. Solving approximately for $\text{var}(y_i)$ in (12) yields

$$\text{var}(y_i) = (1 - h_{ii})^{-2} E(\hat{r}_i^2).$$

Replacing $E(\hat{r}_i^2)$ by the observed $\hat{r}_i^2$ gives the bias-corrected estimator of $\text{var}(y_i)$:

$$\widehat{\text{var}}(y_i) = (1 - h_{ii})^{-2} \hat{r}_i^2.$$

Substituting $\widehat{\text{var}}(y_i)$ into the sandwich form yields the bias-corrected covariance estimator

$$\mathbf{V}_{\lambda,c}^{(1)}(\hat{\boldsymbol{\theta}}) = \phi^{-2} \mathbf{I}_{\lambda}^{-1}(\boldsymbol{\theta}) \left\{ \sum_{i=1}^n \nu_i^{-2} (1 - h_{ii})^{-2} \hat{r}_i^2 \mathbf{G}_i^{\otimes 2} \right\} \mathbf{I}_{\lambda}^{-1}(\boldsymbol{\theta}),$$

where

$$h_{ii} = \frac{1}{\nu_i g'^2(\mu_i)} \mathbf{E}_i^{\top} \mathbf{A} \mathbf{E}_i, \quad \mathbf{A} = \left(\sum_{i=1}^n \nu_i^{-1} \mathbf{G}_i^{\otimes 2} \right)^{-1} = \left\{ \sum_{i=1}^n \frac{1}{\nu_i g'^2(\mu_i)} \mathbf{E}_i^{\otimes 2} \right\}^{-1}.$$

4.3.2 Finite-Sample Correction to the Variance Estimator

Morel et al. (2003) introduced a small sample correction in the Taylor series based estimator of the covariance matrix of the parameter ~~estimates~~ **estimators**. The correction reduces the bias of the Type I error rates in small samples and guarantees positive definiteness of the estimated ~~variance-covariance~~ **covariance** matrix. [I changed “variance-covariance” to “covariance” just for consistency.] To adjust for the bias incurred by the sandwich estimator, Morel et al. (2003) recommended an inflated estimator by adding a scaled version of trace to the sandwich estimator. Incorporating the suggestion by Morel et al. (2003), we proposed an adjusted estimator to atone for the deficiency in variance estimation.

The modified variance-covariance estimator is referred to as

$$\mathbf{V}_{\lambda,c}^{(2)}(\hat{\boldsymbol{\theta}}) = \mathbf{V}_{\lambda,r}(\hat{\boldsymbol{\theta}}) + \delta \kappa \mathbf{I}_{\lambda}^{-1}(\hat{\boldsymbol{\theta}}),$$

where

$$\delta = \min \left(0.5, \frac{\rho}{n - \rho} \right) \quad \text{and} \quad \kappa = \max \left[1, \rho^{-1} \text{trace} \{ \mathbf{I}_{\lambda}^{-1}(\boldsymbol{\theta}) \mathbf{H}(\boldsymbol{\theta}) \} \right].$$

The quantity $\text{trace} \{ \mathbf{I}(\boldsymbol{\theta})^{-1} \mathbf{H}(\boldsymbol{\theta}) \}$ is referred to generalized design effect (Morel et al. 2003). Note that the above bias-corrected variance estimator is guaranteed to be positive definite provided $\mathbf{I}(\boldsymbol{\theta})$ is non-singular.

The Jacobian matrix of the mapping $F : \boldsymbol{\theta} = (\boldsymbol{\beta}^{\top}, \boldsymbol{\eta}^{\top}, \boldsymbol{\zeta}^{\top})^{\top} \rightarrow \boldsymbol{\theta}_{\eta} = (\boldsymbol{\beta}^{\top}, \boldsymbol{\alpha}_{\eta}^{\top}, \boldsymbol{\zeta}^{\top})^{\top}$ is

$$\mathbf{J}(\boldsymbol{\eta}) = \begin{bmatrix} \mathbf{I}_d & \mathbf{0}_{d \times (p-1)} & \mathbf{0}_{d \times (q_n-1)} \\ \mathbf{0}_{1 \times d} & -h^{-3/2} \boldsymbol{\eta}^{\top} & \mathbf{0}_{1 \times (q_n-1)} \\ \mathbf{0}_{(p-1) \times d} & -h^{-3/2} \boldsymbol{\eta}^{\otimes 2} + h^{-1/2} \mathbf{I}_{p-1} & \mathbf{0}_{(p-1) \times (q_n-1)} \\ \mathbf{0}_{(q_n-1) \times d} & \mathbf{0}_{(q_n-1) \times (p-1)} & \mathbf{I}_{q_n-1} \end{bmatrix}$$

where $h = 1 + \|\boldsymbol{\eta}\|^2$. Notice that $\mathbf{J}(\boldsymbol{\eta})$ is a matrix of dimension $(d + p + q_n - 1) \times (d + p - 1 + q_n - 1)$. In our simulation setting (i.e., $p = 3$ and $d = 2$),

$$\mathbf{J}(\boldsymbol{\eta}) = \begin{bmatrix} \mathbf{I}_2 & \mathbf{0}_{2 \times 1} & \mathbf{0}_{2 \times 1} & \mathbf{0}_{2 \times (q_n-1)} \\ \mathbf{0}_{1 \times 2} & -h^{-3/2} \eta_1 & -h^{-3/2} \eta_2 & \mathbf{0}_{1 \times (q_n-1)} \\ \mathbf{0}_{1 \times 2} & h^{-1/2} - h^{-3/2} \eta_1^2 & -h^{-3/2} \eta_1 \eta_2 & \mathbf{0}_{1 \times (q_n-1)} \\ \mathbf{0}_{1 \times 2} & -h^{-3/2} \eta_1 \eta_2 & h^{-1/2} - h^{-3/2} \eta_2^2 & \mathbf{0}_{1 \times (q_n-1)} \\ \mathbf{0}_{(q_n-1) \times 2} & \mathbf{0}_{(q_n-1) \times 1} & \mathbf{0}_{(q_n-1) \times 1} & \mathbf{I}_{q_n-1} \end{bmatrix}$$

Hence, the robust and bias-corrected variance estimators of $\hat{\boldsymbol{\theta}}_{\boldsymbol{\eta}}$ is given by

$$\mathbf{V}(\hat{\boldsymbol{\theta}}_{\boldsymbol{\eta}}) = \mathbf{J}(\boldsymbol{\eta})\mathbf{V}(\hat{\boldsymbol{\theta}})\mathbf{J}^{\top}(\boldsymbol{\eta}),$$

where $\mathbf{V}(\hat{\boldsymbol{\theta}})$ is the generic form of robust and bias-corrected variance estimators for $\hat{\boldsymbol{\theta}}$.

5 Simulation Studies and Real Data Analysis

5.1 Simulation Studies

We conducted extensive simulation studies to evaluate the finite-sample performance of the proposed penalized spline estimator for generalized partially linear single-index models (GPLSIMs). The objectives were to assess (i) the estimation accuracy of the regression and index parameters $(\boldsymbol{\beta}, \boldsymbol{\alpha})$, (ii) the recovery of the nonlinear link function $\varphi(\cdot)$, and (iii) the performance of several variance estimators.

Two canonical generalized linear model settings were considered: a binomial model with a logistic link and a Poisson model with a log link. In both cases, data were generated from the $g\{\mu_i\} = \boldsymbol{\beta}^\top \mathbf{x}_i + \varphi(\boldsymbol{\alpha}^\top \mathbf{z}_i)$, where $\boldsymbol{\alpha} \in \mathbb{R}^p$ is the index parameter, $\boldsymbol{\beta} \in \mathbb{R}^d$ is the vector of regression coefficients, and $\varphi(\cdot)$ is an unknown smooth function. The true parameters were specified as $\boldsymbol{\alpha}_0 = \frac{1}{\sqrt{3}}(1, 1, 1)^\top$, $\boldsymbol{\beta}_0 = (1, -0.5)^\top$, and the nonlinear function was defined as $\varphi(u) = \sin(2u)$. To ensure identifiability, the nonlinear component was empirically centered by subtracting $\mathbb{E}\{\varphi(\boldsymbol{\alpha}^\top \mathbf{z}_i)\}$.

The covariates were generated independently as $Z_1, Z_2, Z_3 \sim \text{Uniform}(0, 2)$, $X_1 \sim N(0, 1)$, $X_2 \sim \text{Bernoulli}(0.5)$. We denote $\mathbf{z}_i = (Z_{i1}, Z_{i2}, Z_{i3})^\top$ and $\mathbf{x}_i = (X_{i1}, X_{i2})^\top$. Poisson outcomes are generated by $Y_i \sim \text{Poisson}(\mu_i)$, $\mu_i = C_i \exp(\boldsymbol{\beta}^\top \mathbf{x}_i + \varphi(\boldsymbol{\alpha}^\top \mathbf{z}_i))$, with an offset $C_i \sim \text{Uniform}(0, 10)$. Binomial outcomes are generated by $Y_i \sim \text{Bernoulli}(\mu_i)$, $\mu_i = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)}$, with $\eta_i = \boldsymbol{\beta}^\top \mathbf{x}_i + \varphi(\boldsymbol{\alpha}^\top \mathbf{z}_i)$. The sample size was either $n = 400$, $n = 800$ or $n = 2000$ with 1000 replications. The unknown function $\varphi(\cdot)$ was approximated using a cubic B-spline basis with a second-order difference penalty. Parameters were estimated jointly via an iterative two-stage algorithm consisting of Fisher scoring with step-halving and smoothing parameter updates using the generalized Fellner–Schall method. The index parameter $\boldsymbol{\alpha}$ was recovered from the reparameterized vector $\boldsymbol{\eta}$ through $\boldsymbol{\alpha} = \frac{(1, \boldsymbol{\eta}^\top)^\top}{\sqrt{1 + \|\boldsymbol{\eta}\|^2}}$.

Tables 1 summarizes the parameter estimation results. In the Poisson model, the proposed estimator (Lu) exhibits negligible bias and substantially smaller variability than the competing method [?] (Yu). For example, at $n = 800$, the standard deviation of α_1 is reduced from 0.0431 under Yu to 0.0123 under Lu, corresponding to a reduction of approximately 71%. Similar efficiency gains are observed for the other index parameters, with Lu producing much smaller SD and MSE values across α_1 , α_2 , and α_3 . The parametric components are also accurately recovered, with very small bias for both β_1 and β_2 . In the Binomial model, estimation is more challenging, especially for the index parameters. Nevertheless, Lu remains more stable than Yu. At $n = 800$, Yu shows noticeable bias for α_2 and α_3 with biases of 0.0716 and 0.0679, respectively, whereas Lu reduces these to 0.0070 and 0.0065. Lu also has considerably smaller variability for the index parameters; for example, the SD of α_2 decreases from 0.2878 under Yu to 0.0839 under Lu. As the sample size increases to $n = 2000$ (See Tables 4–5), both methods improve, but Lu continues to show smaller SD and MSE values for most parameters. Overall, the results indicate that the proposed estimator provides accurate and stable recovery of both the index and parametric components, with the strongest efficiency advantages observed for the index parameters.

We compared four variance estimators: model-based (`se.mod`), robust (`se.rob`), leverage-adjusted (`se.lev`), and finite-sample corrected (`se.fin`). Results are reported in Tables 6 and 7. In the Poisson model, all estimators perform similarly, with empirical standard deviations closely matching estimated standard errors and coverage probabilities near the nominal 95% level. In contrast, the Binomial model shows

Table 1: Summary of parameter estimation for Poisson and Binomial simulation studies.

Parameter	Metric	Poisson				Binomial			
		$n = 400$		$n = 800$		$n = 400$		$n = 800$	
		Lu	Yu	Lu	Yu	Lu	Yu	Lu	Yu
α_1	Mean	0.5767	0.5624	0.5774	0.5734	0.5610	0.5376	0.5724	0.5595
	Bias	0.0006	0.0149	-0.0001	0.0040	0.0164	0.0397	0.0049	0.0178
	SD	0.0176	0.1154	0.0123	0.0431	0.1335	0.2183	0.0834	0.1522
	MSE	0.0003	0.0135	0.0002	0.0019	0.0181	0.0492	0.0070	0.0234
α_2	Mean	0.5774	0.5430	0.5777	0.5787	0.5610	0.3999	0.5704	0.5057
	Bias	-0.0000	0.0343	-0.0003	-0.0013	0.0163	0.1775	0.0070	0.0716
	SD	0.0178	0.1925	0.0125	0.0437	0.1389	0.4216	0.0839	0.2878
	MSE	0.0003	0.0382	0.0002	0.0019	0.0195	0.2091	0.0071	0.0879
α_3	Mean	0.5771	0.5433	0.5765	0.5751	0.5625	0.4059	0.5709	0.5094
	Bias	0.0003	0.0340	0.0008	0.0023	0.0149	0.1715	0.0065	0.0679
	SD	0.0180	0.2081	0.0123	0.0438	0.1310	0.4016	0.0840	0.2566
	MSE	0.0003	0.0444	0.0002	0.0019	0.0174	0.1905	0.0071	0.0704
β_1	Mean	1.0003	1.0026	0.9996	0.9972	1.0170	1.0186	1.0094	1.0072
	Bias	-0.0003	-0.0026	0.0004	0.0028	-0.0170	-0.0186	-0.0094	-0.0072
	SD	0.0166	0.0716	0.0111	0.0520	0.1422	0.1494	0.0985	0.0999
	MSE	0.0003	0.0051	0.0001	0.0027	0.0205	0.0226	0.0098	0.0100
β_2	Mean	-0.5005	-0.5055	-0.4996	-0.5022	-0.5031	-0.5096	-0.5003	-0.5004
	Bias	0.0005	0.0055	-0.0004	0.0022	0.0031	0.0096	0.0003	0.0004
	SD	0.0346	0.1171	0.0235	0.0821	0.1647	0.2364	0.1146	0.1594
	MSE	0.0012	0.0137	0.0006	0.0067	0.0271	0.0559	0.0131	0.0254

$\alpha_0 = (1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3})^\top$ and $\beta_0 = (1, -0.5)^\top$. Mean, bias, standard deviation (SD), and mean squared error (MSE) are based on 1000 Monte Carlo replications.

systematic underestimation of variability by the model-based estimator at small ($n = 400$) or moderate ($n = 800$) sample sizes (See Table 7). The corrected estimators improve coverage, with the leverage-adjusted correction providing the most stable performance. At $n = 2000$, all estimators converge, and coverage approaches the nominal level. These results indicate that variance correction is necessary for valid inference in moderate sample sizes, particularly for nonlinear models.

Figure 1 summarizes the finite-sample performance of the proposed variance estimation procedures for the Poisson and Binomial GPLSIMs across different sample sizes. Panel (A) compares the ratio of the model-based standard error to the empirical standard deviation, where values close to one indicate accurate variance estimation. For the Poisson model, the ratios remain close to one for both the index parameters α and the regression parameters β across all sample sizes, indicating excellent calibration of the model-based variance estimator. In contrast, the Binomial model exhibits noticeable underestimation of variability for the index parameters when the sample size is small, with the ratios substantially below one at $n = 400$ and $n = 800$, although calibration improves as the sample size increases to $n = 2000$. The regression parameters show comparatively stable performance throughout. Panel (B) displays the empirical coverage probabilities of nominal 95% confidence intervals. Consistent with the calibration results, the Poisson model achieves coverage probabilities close to the nominal level for all parameters, whereas the Binomial model exhibits undercoverage for the index parameters in moderate sample sizes due to variance underestimation. Nevertheless, the coverage probabilities steadily approach the target level as

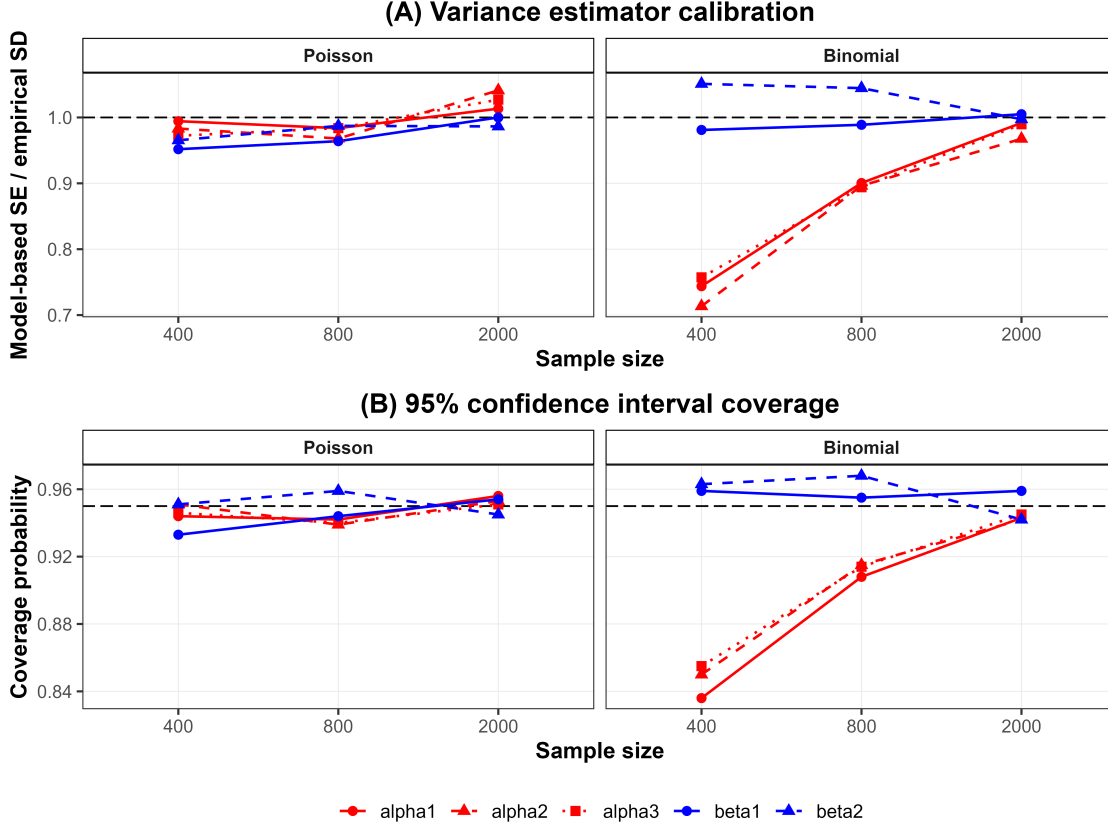


Figure 1: Summary of variance estimation performance for the Poisson and Binomial simulation studies, based on 1,000 Monte Carlo replications. Panel (A) displays the ratio of the model-based standard error to the empirical standard deviation, where the dashed horizontal line at 1 indicates perfect calibration. Panel (B) displays the empirical coverage probability of the nominal 95% confidence intervals, where the dashed horizontal line indicates the target coverage level of 0.95. Red curves correspond to the α parameters and blue curves correspond to the β parameters.

the sample size increases, demonstrating the asymptotic validity of the proposed inference procedure and highlighting the greater finite-sample difficulty associated with estimating nonlinear index parameters in logistic GPLSIMs.

Figure 3 displays the estimated link function along with 95% confidence bands for both Poisson and Binomial models across different sample sizes. Overall, the proposed estimator successfully captures the underlying nonlinear pattern of the true link function $\varphi(u) = \sin(2u)$ in all settings. In the Poisson model (left panels), the estimated curves closely track the true function even at $n = 400$, with only minor deviations near the boundaries. As the sample size increases to $n = 800$ and $n = 2000$, the estimation becomes nearly indistinguishable from the truth, and the confidence bands tighten substantially, indicating strong efficiency and consistency. In contrast, the Binomial model (right panels) exhibits greater variability, particularly at $n = 400$, where noticeable deviations from the true curve and wider confidence bands are observed, especially near the boundaries and regions with sparse data. However, as the sample size increases, the estimator improves markedly. At $n = 800$, the overall shape is well recovered despite some remaining variability, and by $n = 2000$, the estimated curve aligns closely with the true function, with

significantly narrower confidence bands. Across both models, the reduction in band width and improved alignment with the true function as n increases provide clear evidence of consistency. The results also highlight that, while the Binomial setting is inherently more challenging, the proposed estimator remains robust and achieves accurate recovery of the link function with sufficient sample size.

Overall, the proposed estimator achieves accurate parameter estimation, reliable recovery of nonlinear structure, and valid inference. It consistently outperforms the competing method in efficiency and stability, with gains most pronounced for index parameters and in smaller samples. The variance study further highlights the importance of correction methods for achieving nominal coverage in finite samples.

5.2 Real Data Analysis

We illustrate the proposed generalized partially linear single-index model (GPLSIM) using data from the National Health and Nutrition Examination Survey (NHANES), a nationally representative cross-sectional study designed to assess the health and nutritional status of adults and children in the United States [?]. The data are accessed via the `NHANES` R package [?], which provides a curated subset suitable for statistical analysis. We consider a binary diabetes outcome, *Diabetes*. The single-index component includes body mass index (*BMI*), systolic blood pressure (*BPSysAve*), and directly measured HDL cholesterol (*DirectChol*), while the linear component includes age, gender (*Gender*), and race. To improve stability and interpretability, the original *Hispanic* and *Mexican* race categories are combined into a single *Hispanic* category because of their relatively small sample sizes, and the *White* group is used as the reference category in the regression analysis. Continuous predictors are standardized prior to model fitting. We compare three approaches: a parametric generalized linear model (GLM) [?], a nonparametric generalized additive model (GAM) [?], and the proposed GPLSIM (Lu GPLSIM).

Table 2: Comparison of the estimated effects of the single-index variables under the GLM, GAM, and proposed GPLSIM (Lu GPLSIM).

Model	/	BMI	SBP	HDL cholesterol (DirectChol)
Quantity				
GLM				
Estimate	(<i>p</i> -value)	OR = 1.76 (***)	OR = 1.07 (NS)	OR = 0.73 (***)
Interpretation		Significant positive linear effect	No significant linear association	Significant protective linear effect
GAM				
Estimate	(<i>p</i> -value)	EDF = 2.382 (***)	EDF = 3.591 (NS)	EDF = 1.002 (***)
Interpretation		Significant nonlinear smooth effect	No significant smooth effect	Significant approximately linear smooth effect
Lu GPLSIM				
Estimate	(<i>p</i> -value)	$\hat{\alpha}_1 = 0.839$ (***)	$\hat{\alpha}_2 = 0.135$ (**)	$\hat{\alpha}_3 = -0.527$ (***)
Interpretation		Largest positive contributor to the latent index	Small positive contributor to the latent index	Large negative contributor to the latent index

Note: For the GLM, estimates are reported as odds ratios (ORs). For the GAM, estimates are summarized by the effective degrees of freedom (EDF) of the smooth function, where EDF close to one indicates an approximately linear relationship and larger EDF indicates increasing nonlinearity. For the proposed GPLSIM, the estimated coefficients quantify the relative contribution of each variable to the latent single index rather than individual log-odds effects. Significance codes are defined as *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, and NS: not statistically significant ($p \geq 0.05$). Across all three approaches, BMI consistently emerges as the strongest positive contributor to diabetes risk, HDL cholesterol exhibits a protective association, and systolic blood pressure contributes comparatively little after adjustment for the remaining covariates.

The estimated effects of the three single-index variables are remarkably consistent across the GLM, GAM, and the proposed GPLSIM, although the three models represent these effects differently. The GLM summarizes each predictor through a constant odds ratio, the GAM models each predictor by a separate smooth function characterized by its effective degrees of freedom (EDF), and the proposed GPLSIM combines the variables into a common latent metabolic index whose coefficients quantify their relative contributions and directions rather than individual odds ratios.

For body mass index (BMI), the GLM indicates a strong positive association with diabetes, with an estimated odds ratio of 1.76 ($p < 0.001$). The GAM likewise identifies BMI as a highly significant nonlinear predictor ($\text{EDF} = 2.382$, $p < 0.001$), indicating that its effect cannot be adequately represented by a simple linear relationship. Consistent with these findings, the proposed GPLSIM assigns BMI the largest positive index coefficient ($\hat{\alpha}_1 = 0.839$, $p < 0.001$), indicating that BMI is the dominant contributor to the latent metabolic index. Thus, all three approaches consistently identify BMI as the strongest metabolic risk factor for diabetes.

For systolic blood pressure (SBP), the GLM estimates an odds ratio of 1.07, which is not statistically significant ($p = 0.136$), suggesting little independent association with diabetes after adjusting for the remaining covariates. Similarly, the GAM reports a nonsignificant smooth effect ($\text{EDF} = 3.591$, $p = 0.147$), providing no evidence of a meaningful nonlinear relationship. In contrast, the GPLSIM assigns SBP a small positive index coefficient ($\hat{\alpha}_2 = 0.135$, $p < 0.01$). Although this coefficient is statistically significant, its magnitude is substantially smaller than that of BMI and HDL cholesterol, indicating that SBP contributes relatively little to the construction of the latent metabolic index. Therefore, despite its statistical significance within the index formulation, SBP appears to play only a minor role in the overall metabolic risk profile.

For directly measured HDL cholesterol (DirectChol), the GLM estimates an odds ratio of 0.73 ($p < 0.001$), indicating a significant protective association with diabetes. The GAM likewise identifies a highly significant smooth effect with an effective degree of freedom close to one ($\text{EDF} = 1.002$, $p < 0.001$), implying an approximately linear protective relationship. The proposed GPLSIM similarly assigns HDL cholesterol a large negative index coefficient ($\hat{\alpha}_3 = -0.527$, $p < 0.001$), indicating that increasing HDL levels reduce the latent metabolic index. Assuming a monotone increasing link function, a lower latent index corresponds to a lower probability of diabetes, consistent with the protective effect observed in both the GLM and GAM. Moreover, the magnitude of the GPLSIM coefficient suggests that HDL cholesterol is the second most influential component of the latent metabolic index after BMI.

Overall, the three modeling approaches produce highly consistent substantive conclusions regarding the direction and relative importance of the metabolic risk factors. BMI exhibits the strongest positive contribution to diabetes risk, HDL cholesterol exhibits a substantial protective contribution, and systolic blood pressure contributes comparatively little after accounting for the remaining variables. The principal distinction among the models lies in their representation of these effects: the GLM expresses them through constant odds ratios, the GAM through flexible smooth functions, and the proposed GPLSIM through a latent single index whose coefficients quantify the relative importance of the metabolic variables.

The ability to summarize multiple correlated metabolic risk factors through a single latent index is particularly advantageous in practice. Rather than interpreting several individual nonlinear effects separately, clinicians and public health researchers can evaluate an individual’s overall metabolic status using a single composite score that integrates the joint contributions of BMI, systolic blood pressure, and HDL cholesterol. This lower-dimensional representation facilitates risk stratification, visualization, and prediction while mitigating multicollinearity among correlated biomarkers. Furthermore, because the latent index is estimated directly from the data, it provides an interpretable summary measure of overall metabolic risk that can be incorporated into downstream analyses or clinical decision-making without requiring separate assessment of each nonlinear covariate effect. Consequently, the proposed GPLSIM offers a parsimonious yet flexible framework that preserves the essential nonlinear structure of the metabolic variables while

yielding a clinically meaningful summary of diabetes risk.

Table 3: Comparison of the estimated coefficients for the linear covariates across the GLM, GAM, and GPLSIM.

Covariate	GLM Estimate	GAM Estimate	GPLSIM Estimate
Age (standardized)	1.216***	1.206***	1.212***
Male (vs. Female)	0.177	0.185	0.164
Black (vs. White)	0.778***	0.785***	0.822***
Hispanic (vs. White)	0.664***	0.655***	0.654***
Other (vs. White)	0.842***	0.847***	0.854***

Note: The proposed GPLSIM models age, gender, and race/ethnicity through a linear component, so their coefficients retain the same log-odds interpretation as those from a conventional logistic regression model and may be exponentiated to obtain adjusted odds ratios. White serves as the reference category for race/ethnicity, and the Hispanic category combines the original Mexican American and Other Hispanic groups. The estimated GPLSIM coefficients ($\hat{\alpha}$) are directly comparable with the corresponding GLM and GAM coefficients, facilitating assessment of the consistency of the linear effects across the three competing models. Significance codes: *** $p < 0.001$.

Unlike the single-index variables, whose effects are summarized through a latent nonlinear index, the linear covariates in the proposed GPLSIM retain the same interpretation as those in a conventional logistic regression model. Consequently, their estimated coefficients can be directly compared with those obtained from the GLM and the parametric component of the GAM. As shown in Table 3, the estimated coefficients are remarkably consistent across the three modeling approaches. Age exhibits a strong positive association with diabetes risk in all models, with estimated coefficients ranging from 1.206 to 1.216, indicating that older individuals are substantially more likely to have diabetes after adjusting for the remaining covariates. The estimated gender effect is also highly stable, with positive coefficients between 0.164 and 0.185, suggesting that males tend to have a higher probability of diabetes than females, although the corresponding effect is not statistically significant at the conventional 5% level.

The estimated race effects likewise demonstrate excellent agreement across the three models. Using White as the reference category, Black individuals exhibit a significantly higher risk of diabetes, with estimated coefficients ranging from 0.778 to 0.822. Similarly, the combined Hispanic category, which merges the original Mexican American and Other Hispanic groups, shows a significantly elevated diabetes risk relative to White individuals, with estimated coefficients around 0.65 across all models. The Other race category also exhibits a consistently positive association with diabetes, with estimated coefficients between 0.842 and 0.854. The close numerical agreement of these estimates demonstrates that the proposed GPLSIM preserves the interpretation of the demographic covariates while allowing greater modeling flexibility for the metabolic variables represented through the single-index component.

Because age, gender, and race enter the GPLSIM through a linear component, their coefficients retain the standard log-odds interpretation and can be exponentiated to obtain adjusted odds ratios exactly as in a conventional logistic regression model. This property distinguishes the GPLSIM from a fully nonparametric additive model, where nonlinear smooth functions are estimated for all continuous variables and direct odds-ratio interpretations are generally unavailable. Consequently, the proposed GPLSIM combines the interpretability of classical logistic regression for demographic covariates with the flexibility of a nonlinear latent-index representation for the correlated metabolic predictors, yielding a parsimonious yet scientifically interpretable framework for diabetes risk assessment. The high level of agreement between the GPLSIM and the corresponding GLM and GAM estimates further indicates that incorporating a nonlinear latent metabolic index does not materially alter inference for the linear demographic covariates

while substantially enriching the representation of the correlated metabolic risk factors.

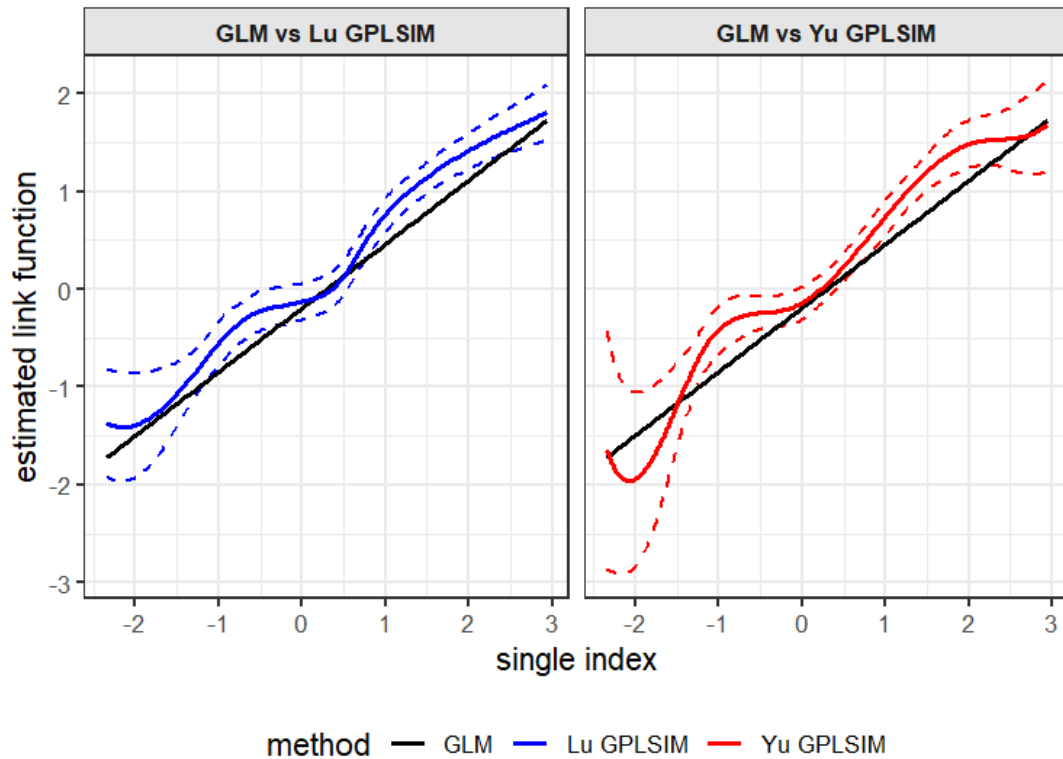


Figure 2: Estimated link functions along the single-index direction for the binary outcome of diabetes (*Diabetes*). The left panel compares the GLM (black) with the Lu GPLSIM estimator (blue), while the right panel compares the GLM (black) with the Yu GPLSIM estimator (red). Dashed curves denote pointwise 95% confidence intervals for the estimated nonparametric link functions.

Figure 2 presents the estimated link functions along the single-index direction for the binary diabetes outcome. The left panel compares the GLM and the Lu GPLSIM estimator, while the right panel compares the GLM and the Yu GPLSIM estimator. The GLM imposes a strictly linear relationship between the single index and the log-odds of diabetes, whereas both GPLSIM approaches estimate the link function nonparametrically. Overall, the two GPLSIM estimators reveal a nonlinear monotone increasing relationship between the latent metabolic index and diabetes risk that is not fully captured by the linear GLM.

The Lu GPLSIM estimate exhibits a smooth S-shaped pattern, with a relatively gradual increase for smaller index values, followed by a steeper rise in the middle region before approaching an approximately linear trend for larger index values. The estimated confidence bands remain relatively narrow across most of the index domain, indicating stable estimation except near the boundaries where observations are sparse. In comparison, the Yu GPLSIM produces a qualitatively similar nonlinear pattern but displays stronger curvature in the lower-index region and somewhat wider confidence intervals, reflecting greater variability in estimating the link function near the extremes of the observed index values. Despite these local differences, both GPLSIM approaches consistently suggest that the association between the latent

metabolic score and diabetes risk deviates from the linear relationship assumed by the GLM.

Overall, these results demonstrate the advantage of generalized partially linear single-index models for modeling complex metabolic effects on diabetes risk. By combining multiple correlated biomarkers into a single latent metabolic index while estimating its nonlinear effect nonparametrically, both GPLSIM methods provide substantially greater flexibility than the conventional GLM. Moreover, the close agreement between the Lu and Yu GPLSIM estimators indicates that the inferred nonlinear relationship is robust across estimation procedures, while the Lu GPLSIM yields a smoother estimate with narrower confidence bands over much of the index range.

6 Technical Details

6.1 Common Single-index models

In this section we discuss the estimation of common single-index models, including partially linear single-index models, partially linear logistic single-index models, and partially linear Poisson single-index models.

Partially Linear Logistic Single-Index Models. Consider partially linear single-index models

$$y_i = \mu_i + \varepsilon_i = \boldsymbol{\beta}^\top \mathbf{x}_i + \boldsymbol{\zeta}^\top \mathbf{b}_i + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2) \quad (13)$$

The penalized log-likelihood of $\boldsymbol{\theta}$ is given by

$$l_\lambda(\boldsymbol{\theta}) = -\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu_i)^2 - \frac{1}{2\sigma^2} \lambda \boldsymbol{\zeta}^\top \mathbf{D}^\top \mathbf{D} \boldsymbol{\zeta}.$$

In the following are gradient and Fisher of $l_\lambda(\boldsymbol{\theta})$ and robust variance estimators of $\hat{\boldsymbol{\theta}}$:

- The gradient of $l_\lambda(\boldsymbol{\theta})$ is

$$\nabla l_\lambda(\boldsymbol{\theta}) = 1/\sigma^2 \mathbf{E}^\top (\mathbf{y} - \boldsymbol{\mu}) - \lambda/\sigma^2 \mathbf{P} \boldsymbol{\theta}.$$

- The Fisher information of $l_\lambda(\boldsymbol{\theta})$ is

$$\mathbf{I}_\lambda(\boldsymbol{\theta}) = 1/\sigma^2 \mathbf{E}^\top \mathbf{E} + \lambda/\sigma^2 \mathbf{P}.$$

- For model (13), $g(\mu_i) = \mu_i$, $V(\mu_i) = 1$, and $\phi = \sigma^2$. Thus, $\mathbf{G} = \mathbf{E}$ and $\mathbf{V}$ is an identity matrix. Further, σ^2 can be updated via an adjusted Pearson statistics methods, namely,

$$\hat{\sigma}^2 = \frac{1}{\rho} \sum_{i=1}^n (y_i - \hat{\mu}_i)^2,$$

where ρ is an adjusted degrees of freedom defined as $\text{tr} \left\{ (\mathbf{E}^\top \mathbf{E} + \lambda \mathbf{P})^{-1} \mathbf{E}^\top \mathbf{E} \right\}$.

- For robust variance estimation, $U_i(\boldsymbol{\theta}) = 1/\sigma^2 (y_i - \mu) \mathbf{E}_i$, and hence,

$$\mathbf{H}(\boldsymbol{\theta}) = \sum_{i=1}^n \text{var}(U_i(\boldsymbol{\theta})) U_i(\boldsymbol{\theta}) U_i(\boldsymbol{\theta})^\top = 1/\sigma^4 \mathbf{E}^\top \mathbf{F} \mathbf{E},$$

where $\mathbf{F} = \text{diag}\{(y_1 - \mu_1)^2, \dots, (y_n - \mu_n)^2\}$. $\mathbf{F} = \text{diag}\{\text{var}(y_1 - \mu_1), \dots, \text{var}(y_n - \mu_n)\} = \sigma^2 \mathbf{I}$.

- The robust variance of $\hat{\boldsymbol{\theta}}$ can be written as

$$\mathbf{V}_{\lambda,r}(\hat{\boldsymbol{\theta}}) = \mathbf{I}_\lambda^{-1}(\hat{\boldsymbol{\theta}}) \mathbf{H}(\hat{\boldsymbol{\theta}}) \mathbf{I}_\lambda^{-1}(\hat{\boldsymbol{\theta}}) = \mathbf{I}_\lambda^{-1}(\boldsymbol{\theta}) \mathbf{H}(\boldsymbol{\theta}) \mathbf{I}_\lambda^{-1}(\boldsymbol{\theta}) = (\mathbf{E}^\top \mathbf{E} + \lambda \mathbf{P})^{-1} (\mathbf{E}^\top \mathbf{F} \mathbf{E}) (\mathbf{E}^\top \mathbf{E} + \lambda \mathbf{P})^{-1}.$$

- For leverage-adjusted robust variance estimation,

$$\mathbf{V}_{\lambda,c}^{(1)}(\hat{\boldsymbol{\theta}}) = (\mathbf{E}\mathbf{E}^\top + \lambda\mathbf{P})^{-1} \left\{ \sum_{i=1}^n \left(\frac{y_i - \hat{\mu}_i}{1 - h_{ii}} \right)^2 \mathbf{E}_i^{\otimes 2} \right\} (\mathbf{E}\mathbf{E}^\top + \lambda\mathbf{P})^{-1},$$

where $h_{ii} = \mathbf{E}_i^\top \mathbf{A} \mathbf{E}_i$ with $\mathbf{A} = \left(\sum_{i=1}^n \mathbf{E}_i^{\otimes 2} \right)^{-1}$,

- The sample correction robust variance estimator is given by

$$\mathbf{V}_{\lambda,c}^{(2)}(\hat{\boldsymbol{\theta}}) = \mathbf{V}_{\lambda,r}(\hat{\boldsymbol{\theta}}) + \kappa \delta (\mathbf{E}\mathbf{E}^\top + \lambda\mathbf{P})^{-1},$$

where $\delta = \min \{0.5, \rho/(n - \rho)\}$ and $\kappa = \max [1, \text{trace} \{(\mathbf{E}\mathbf{E}^\top + \lambda\mathbf{P})^{-1} \mathbf{H}(\boldsymbol{\theta})\}]$.

$$\begin{aligned} \kappa &= \max [1, \rho^{-1} \text{trace} \{ \mathbf{I}_\lambda^{-1}(\boldsymbol{\theta}) \mathbf{H}(\boldsymbol{\theta}) \}] \\ &= \max [1, \rho^{-1} \text{trace} \{ (1/\sigma^2 \mathbf{E}\mathbf{E}^\top + \lambda/\sigma^2 \mathbf{P})^{-1} \mathbf{H}(\boldsymbol{\theta}) \}] \\ &= \max [1, \rho^{-1} \sigma^2 \text{trace} \{ (\mathbf{E}\mathbf{E}^\top + \lambda\mathbf{P})^{-1} \mathbf{H}(\boldsymbol{\theta}) \}] \end{aligned}$$

Partially Linear Logistic Single-Index Models. Consider partially linear logistic single-index models

$$\log \frac{\mu_i}{1 - \mu_i} = \boldsymbol{\beta}^\top \mathbf{z}_i + \boldsymbol{\zeta}^\top \mathbf{b}_i \iff \mu_i = \frac{\exp(\boldsymbol{\beta}^\top \mathbf{z}_i + \boldsymbol{\zeta}^\top \mathbf{b}_i)}{1 + \exp(\boldsymbol{\beta}^\top \mathbf{z}_i + \boldsymbol{\zeta}^\top \mathbf{b}_i)}. \quad (14)$$

The penalized log-likelihood of $\boldsymbol{\theta}$ is given by

$$l_\lambda(\boldsymbol{\theta}) = \sum_{i=1}^n \{y_i \log \mu_i + (1 - y_i) \log (1 - \mu_i)\} - \frac{1}{2} \lambda \boldsymbol{\zeta}^\top \mathbf{D}^\top \mathbf{D} \boldsymbol{\zeta}.$$

In the following are gradient and Fisher of $l_\lambda(\boldsymbol{\theta})$ and robust variance estimators of $\hat{\boldsymbol{\theta}}$:

- The gradient of $l_\lambda(\boldsymbol{\theta})$ is

$$\nabla l_\lambda(\boldsymbol{\theta}) = \mathbf{E}^\top (\mathbf{y} - \boldsymbol{\mu}) - \lambda \mathbf{P} \boldsymbol{\theta}$$

- The Fisher information of $l_\lambda(\boldsymbol{\theta})$ is

$$\mathbf{I}_\lambda(\boldsymbol{\theta}) = \mathbf{E}^\top \mathbf{V}_1 \mathbf{E} + \lambda \mathbf{P},$$

where $\mathbf{V}_1 = \text{diag} \{ \mu_1(1 - \mu_1), \dots, \mu_n(1 - \mu_n) \}$.

- For model (14), $g(\mu_i) = \log \frac{\mu_i}{1 - \mu_i}$, $V(\mu_i) = \mu_i(1 - \mu_i)$, and $\phi = 1$. Thus, $\mathbf{G} = \mathbf{V}_1 \mathbf{E}$.
- For robust variance estimation, $U_i(\boldsymbol{\theta}) = (y_i - \mu_i) \mathbf{E}_i$, and hence

$$\mathbf{H}(\boldsymbol{\theta}) = \sum_{i=1}^n U_i(\boldsymbol{\theta})^{\otimes 2} \text{var}(U_i(\boldsymbol{\theta})) = \mathbf{E}^\top \mathbf{F} \mathbf{E},$$

where $\mathbf{F} = \text{diag} \{ \text{var}(y_1 - \mu_1), \dots, \text{var}(y_n - \mu_n) \} = \text{diag} \{ \mu_1(1 - \mu_1), \dots, \mu_n(1 - \mu_n) \} = \mathbf{V}_1$.

- The robust variance of $\widehat{\boldsymbol{\theta}}$ can be written as

$$\mathbf{V}_{\lambda,r}(\widehat{\boldsymbol{\theta}}) = \cancel{\mathbf{I}_{\lambda}^{-1}(\widehat{\boldsymbol{\theta}})\mathbf{H}(\widehat{\boldsymbol{\theta}})\mathbf{I}_{\lambda}^{-1}(\widehat{\boldsymbol{\theta}})} = \mathbf{I}_{\lambda}^{-1}(\boldsymbol{\theta})\mathbf{H}(\boldsymbol{\theta})\mathbf{I}_{\lambda}^{-1}(\boldsymbol{\theta}) = (\mathbf{E}^{\top}\mathbf{V}_1\mathbf{E} + \lambda\mathbf{P})^{-1}(\mathbf{E}^{\top}\mathbf{F}\mathbf{E})(\mathbf{E}^{\top}\mathbf{V}_1\mathbf{E} + \lambda\mathbf{P})^{-1}.$$

- For leverage-adjusted robust variance estimation,

$$\mathbf{V}_{\lambda,c}^{(1)}(\widehat{\boldsymbol{\theta}}) = (\mathbf{E}\mathbf{V}_1\mathbf{E}^{\top} + \lambda\mathbf{P})^{-1} \left\{ \sum_{i=1}^n \left(\frac{y_i - \widehat{\mu}_i}{1 - h_{ii}} \right)^2 \mathbf{E}_i^{\otimes 2} \right\} (\mathbf{E}\mathbf{V}_1\mathbf{E}^{\top} + \lambda\mathbf{P})^{-1},$$

$$\text{where } h_{ii} = \mu_i(1 - \mu_i)\mathbf{E}_i^{\top}\mathbf{A}\mathbf{E}_i \text{ with } \mathbf{A} = \left(\sum_{i=1}^n \mu_i(1 - \mu_i)\mathbf{E}_i^{\otimes 2} \right)^{-1}.$$

- The sample correction robust variance estimator is given by

$$\mathbf{V}_{\lambda,c}^{(2)}(\widehat{\boldsymbol{\theta}}) = \mathbf{V}_{\lambda,r}(\widehat{\boldsymbol{\theta}}) + \delta\kappa(\mathbf{E}\mathbf{V}_1\mathbf{E}^{\top} + \lambda\mathbf{P})^{-1},$$

$$\text{where } \delta = \min\{0.5, \rho/(n - \rho)\} \text{ and } \kappa = \max[1, \text{trace}\{(\mathbf{E}\mathbf{V}_1\mathbf{E}^{\top} + \lambda\mathbf{P})^{-1}\mathbf{H}(\boldsymbol{\theta})\}].$$

$$\begin{aligned} \kappa &= \max[1, \rho^{-1}\text{trace}\{\mathbf{I}_{\lambda}^{-1}(\boldsymbol{\theta})\mathbf{H}(\boldsymbol{\theta})\}] \\ &= \max[1, \rho^{-1}\text{trace}\{(\mathbf{E}\mathbf{V}_1\mathbf{E}^{\top} + \lambda\mathbf{P})^{-1}\mathbf{H}(\boldsymbol{\theta})\}] \end{aligned}$$

Partially Linear Poisson Single-Index Models. Consider partially linear Poisson single-index models

$$\log \mu_i = \log C_i + \boldsymbol{\beta}^{\top}\mathbf{z}_i + \boldsymbol{\zeta}^{\top}\mathbf{b}_i \iff \mu_i = C_i \exp(\boldsymbol{\beta}^{\top}\mathbf{z}_i + \boldsymbol{\zeta}^{\top}\mathbf{b}_i). \quad (15)$$

The penalized log-likelihood of $\boldsymbol{\theta}$ is given by

$$l_{\lambda}(\boldsymbol{\theta}) = \sum_{i=1}^n (y_i \log \mu_i - \mu_i) - \frac{1}{2} \lambda \boldsymbol{\zeta}^{\top} \mathbf{D}^{\top} \mathbf{D} \boldsymbol{\zeta}.$$

In the following are gradient and Fisher of $l_{\lambda}(\boldsymbol{\theta})$ and robust variance estimators of $\widehat{\boldsymbol{\theta}}$:

- The gradient of $l_{\lambda}(\boldsymbol{\theta})$ is

$$\nabla l_{\lambda}(\boldsymbol{\theta}) = \mathbf{E}^{\top}(\mathbf{y} - \boldsymbol{\mu}) - \lambda\mathbf{P}\boldsymbol{\theta}$$

- The Fisher information of $l_{\lambda}(\boldsymbol{\theta})$ is

$$\mathbf{I}_{\lambda}(\boldsymbol{\theta}) = \mathbf{E}^{\top}\mathbf{V}_2\mathbf{E} + \lambda\mathbf{P},$$

where $\mathbf{V}_2 = \text{diag}\{\mu_1, \dots, \mu_n\}$.

- For model (15), $g(\mu_i) = \log \mu_i$, $V(\mu_i) = \mu_i$, and $\phi = 1$. Thus, $\mathbf{G} = \mathbf{V}_2\mathbf{E}$.
- For robust variance estimation, $U_i(\boldsymbol{\theta}) = (y_i - \mu_i)\mathbf{E}_i$, and hence

$$\mathbf{H}(\boldsymbol{\theta}) = \sum_{i=1}^n U_i(\boldsymbol{\theta})^{\otimes 2} = \sum_{i=1}^n (y_i - \mu_i)^2 \mathbf{E}_i^{\otimes 2} = \sum_{i=1}^n \text{var}(U_i(\boldsymbol{\theta})) = \sum_{i=1}^n \text{var}(y_i - \mu_i) \mathbf{E}_i^{\otimes 2} = \mathbf{E}^{\top}\mathbf{F}\mathbf{E},$$

where $\mathbf{F} = \text{dig}\{\text{var}(y_1 - \mu_1), \dots, \text{var}(y_n - \mu_n)\} = \text{dig}\{\mu_1, \dots, \mu_n\} = \mathbf{V}_2$

- The robust variance of $\hat{\boldsymbol{\theta}}$ can be written as

$$\mathbf{V}_{\lambda,r}(\hat{\boldsymbol{\theta}}) = \underline{\mathbf{I}_{\lambda}(\hat{\boldsymbol{\theta}})^{-1} \mathbf{H}(\hat{\boldsymbol{\theta}}) \mathbf{I}_{\lambda}^{-1}(\hat{\boldsymbol{\theta}})} = \mathbf{I}_{\lambda}(\boldsymbol{\theta})^{-1} \mathbf{H}(\boldsymbol{\theta}) \mathbf{I}_{\lambda}^{-1}(\boldsymbol{\theta}) = (\mathbf{E}^{\top} \mathbf{V}_2 \mathbf{E} + \lambda \mathbf{P})^{-1} (\mathbf{E}^{\top} \mathbf{F} \mathbf{E}) (\mathbf{E}^{\top} \mathbf{V}_2 \mathbf{E} + \lambda \mathbf{P})^{-1}.$$

- For leverage-adjusted robust variance estimation,

$$\mathbf{V}_{\lambda,c}^{(1)}(\hat{\boldsymbol{\theta}}) = (\mathbf{E} \mathbf{V}_2 \mathbf{E}^{\top} + \lambda \mathbf{P})^{-1} \left\{ \sum_{i=1}^n \left(\frac{y_i - \hat{\mu}_i}{1 - h_{ii}} \right)^2 \mathbf{E}_i^{\otimes 2} \right\} (\mathbf{E} \mathbf{V}_2 \mathbf{E}^{\top} + \lambda \mathbf{P})^{-1},$$

where $h_{ii} = \mu_i \mathbf{E}_i^{\top} \mathbf{A} \mathbf{E}_i$ with $\mathbf{A} = \left(\sum_{i=1}^n \mu_i \mathbf{E}_i^{\otimes 2} \right)^{-1}$

- The sample correction robust variance estimator is given by

$$\mathbf{V}_{\lambda,c}^{(2)}(\hat{\boldsymbol{\theta}}) = \mathbf{V}_{\lambda,r}(\hat{\boldsymbol{\theta}}) + \delta \kappa (\mathbf{E} \mathbf{V}_2 \mathbf{E}^{\top} + \lambda \mathbf{P})^{-1},$$

where $\delta = \min\{0.5, \rho/(n - \rho)\}$ and $\kappa = \max[1, \rho^{-1} \text{trace}\{(\mathbf{E} \mathbf{V}_2 \mathbf{E}^{\top} + \lambda \mathbf{P})^{-1} \mathbf{H}(\boldsymbol{\theta})\}]$.

6.2 Appendix: Simulation results

Table 4: Summary of parameter estimation for a Poisson simulation study.

Parameter	Metric	Lu			Yu		
		$n = 400$	$n = 800$	$n = 2000$	$n = 400$	$n = 800$	$n = 2000$
α_1	Mean	0.5767	0.5774	0.5774	0.5624	0.5734	0.5743
	Bias	0.0006	-0.0001	0.0000	0.0149	0.0040	0.0031
	SD	0.0176	0.0123	0.0075	0.1154	0.0431	0.0247
	MSE	0.0003	0.0002	0.0001	0.0135	0.0019	0.0006
α_2	Mean	0.5774	0.5777	0.5774	0.5430	0.5787	0.5790
	Bias	-0.0000	-0.0003	0.0000	0.0343	-0.0013	-0.0016
	SD	0.0178	0.0125	0.0073	0.1925	0.0437	0.0250
	MSE	0.0003	0.0002	0.0001	0.0382	0.0019	0.0006
α_3	Mean	0.5771	0.5765	0.5772	0.5433	0.5751	0.5772
	Bias	0.0003	0.0008	0.0002	0.0340	0.0023	0.0002
	SD	0.0180	0.0123	0.0074	0.2081	0.0438	0.0246
	MSE	0.0003	0.0002	0.0001	0.0444	0.0019	0.0006
β_1	Mean	1.0003	0.9996	1.0001	1.0026	0.9972	0.9995
	Bias	-0.0003	0.0004	-0.0001	-0.0026	0.0028	0.0005
	SD	0.0166	0.0111	0.0065	0.0716	0.0520	0.0355
	MSE	0.0003	0.0001	0.0000	0.0051	0.0027	0.0013
β_2	Mean	-0.5005	-0.4996	-0.4998	-0.5055	-0.5022	-0.5019
	Bias	0.0005	-0.0004	-0.0002	0.0055	0.0022	0.0019
	SD	0.0346	0.0235	0.0147	0.1171	0.0821	0.0530
	MSE	0.0012	0.0006	0.0002	0.0137	0.0067	0.0028

$\alpha_0 = (1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3})^\top$ and $\beta_0 = (1, -0.5)$. Sample mean (Mean), bias, standard deviation (SD), and mean squared error (MSE) based on 1000 Monte Carlo samples.

Table 5: Summary of parameter estimation for a Binomial simulation study.

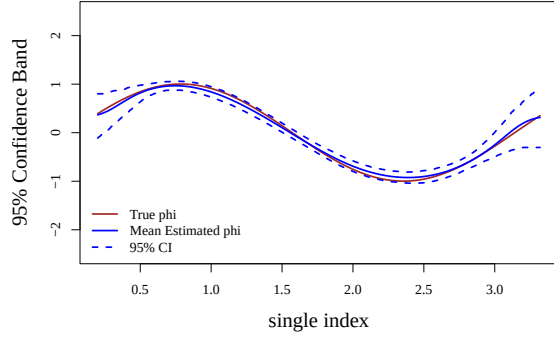
Parameter	Metric	Lu			Yu		
		$n = 400$	$n = 800$	$n = 2000$	$n = 400$	$n = 800$	$n = 2000$
α_1	Mean	0.5610	0.5724	0.5744	0.5376	0.5595	0.5703
	Bias	0.0164	0.0049	0.0029	0.0397	0.0178	0.0070
	SD	0.1335	0.0834	0.0484	0.2183	0.1522	0.0715
	MSE	0.0181	0.0070	0.0023	0.0492	0.0234	0.0052
α_2	Mean	0.5610	0.5704	0.5765	0.3999	0.5057	0.5696
	Bias	0.0163	0.0070	0.0009	0.1775	0.0716	0.0077
	SD	0.1389	0.0839	0.0494	0.4216	0.2878	0.1074
	MSE	0.0195	0.0071	0.0024	0.2091	0.0879	0.0116
α_3	Mean	0.5625	0.5709	0.5750	0.4059	0.5094	0.5669
	Bias	0.0149	0.0065	0.0024	0.1715	0.0679	0.0104
	SD	0.1310	0.0840	0.0484	0.4016	0.2566	0.1107
	MSE	0.0174	0.0071	0.0023	0.1905	0.0704	0.0124
β_1	Mean	1.0170	1.0094	1.0050	1.0186	1.0072	1.0040
	Bias	-0.0170	-0.0094	-0.0050	-0.0186	-0.0072	-0.0040
	SD	0.1422	0.0985	0.0609	0.1494	0.0999	0.0612
	MSE	0.0205	0.0098	0.0037	0.0226	0.0100	0.0038
β_2	Mean	-0.5031	-0.5003	-0.5002	-0.5096	-0.5004	-0.4998
	Bias	0.0031	0.0003	0.0002	0.0096	0.0004	-0.0002
	SD	0.1647	0.1146	0.0748	0.2364	0.1594	0.1042
	MSE	0.0271	0.0131	0.0056	0.0559	0.0254	0.0108

$\alpha_0 = (1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3})^\top$ and $\beta_0 = (1, -0.5)$. Sample mean (Mean), bias, standard deviation (SD), and mean squared error (MSE) based on 1000 Monte Carlo samples.

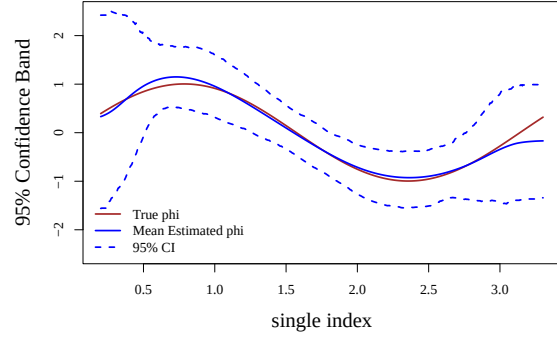
Table 6: Results of variance study for a Poisson simulation.

Model (Sample Size)	Parameter	emp.sd	se.mod	se.rob	se.lev	se.fin	cov.prob.mod
Poisson ($n = 400$)	α_1	0.0176	0.0175	0.0167	0.0181	0.0169	0.944
	α_2	0.0178	0.0175	0.0168	0.0181	0.0170	0.951
	α_3	0.0180	0.0175	0.0167	0.0181	0.0169	0.946
	β_1	0.0166	0.0158	0.0145	0.0168	0.0147	0.933
	β_2	0.0346	0.0334	0.0323	0.0346	0.0327	0.951
Poisson ($n = 800$)	α_1	0.0123	0.0121	0.0118	0.0124	0.0119	0.942
	α_2	0.0125	0.0121	0.0118	0.0124	0.0119	0.939
	α_3	0.0123	0.0121	0.0118	0.0124	0.0119	0.940
	β_1	0.0111	0.0107	0.0102	0.0112	0.0103	0.944
	β_2	0.0235	0.0232	0.0227	0.0236	0.0229	0.959
Poisson ($n = 2000$)	α_1	0.0075	0.0076	0.0075	0.0076	0.0075	0.956
	α_2	0.0073	0.0076	0.0075	0.0076	0.0075	0.953
	α_3	0.0074	0.0076	0.0075	0.0076	0.0075	0.951
	β_1	0.0065	0.0065	0.0064	0.0067	0.0064	0.954
	β_2	0.0147	0.0145	0.0144	0.0146	0.0144	0.945

Empirical standard deviation (emp.sd), model-based standard error (se.mod), robust sandwich standard error (se.rob), leverage-adjusted standard error (se.lev), finite-sample corrected standard error (se.fin), and coverage probability of the 95% confidence interval based on the model-based variance estimator (cov.prob.mod) for (α, β) . Results are based on 1000 Monte Carlo samples with sample sizes $n = 400$, $n = 800$, and $n = 2000$.



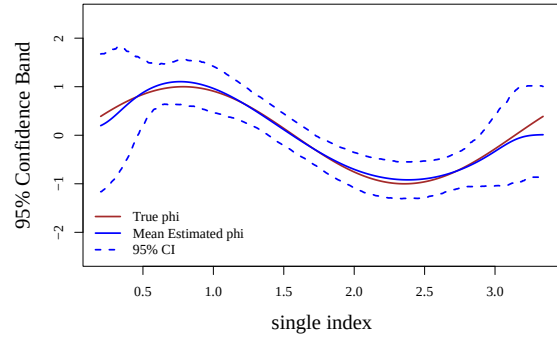
(a) Poisson model ($n = 400$)



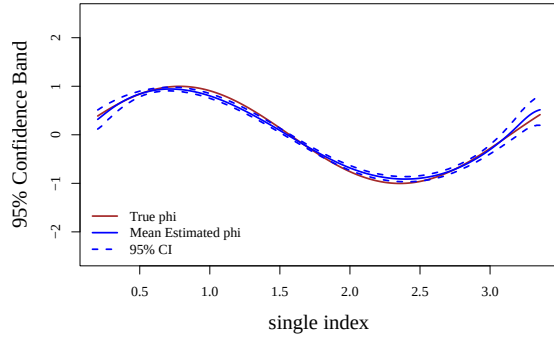
(b) Binomial model ($n = 400$)



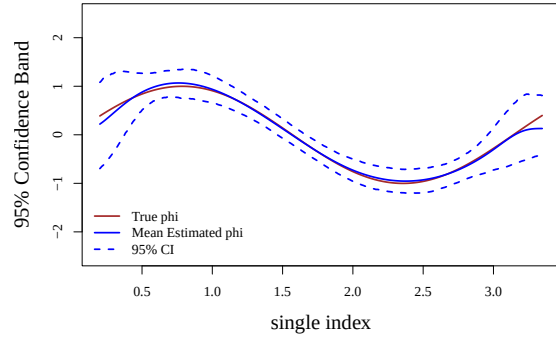
(c) Poisson model ($n = 800$)



(d) Binomial model ($n = 800$)



(e) Poisson model ($n = 2000$)



(f) Binomial model ($n = 2000$)

Figure 3: **Estimated Link Function.** Comparison of the true link function with the mean estimated link function and its pointwise 95% confidence bands for Poisson and Binomial models. The left panels correspond to the Poisson model, while the right panels correspond to the Binomial model. Increasing the sample size from $n = 400$, $n = 800$, to $n = 2000$ results in improved estimation accuracy and narrower confidence bands, illustrating the consistency and efficiency of the proposed method.

Table 7: Results of variance study for a Binomial simulation.

Model (Sample Size)	Parameter	emp.sd	se.mod	se.rob	se.lev	se.fin	cov.prob.mod
Binomial ($n = 400$)	α_1	0.1335	0.0993	0.0984	0.1011	0.0993	0.836
	α_2	0.1389	0.0991	0.0984	0.1013	0.0993	0.850
	α_3	0.1310	0.0992	0.0984	0.1013	0.0994	0.855
	β_1	0.1422	0.1395	0.1387	0.1415	0.1401	0.959
	β_2	0.1647	0.1731	0.1718	0.1760	0.1735	0.963
Binomial ($n = 800$)	α_1	0.0834	0.0751	0.0748	0.0759	0.0752	0.908
	α_2	0.0839	0.0752	0.0749	0.0760	0.0753	0.915
	α_3	0.0840	0.0751	0.0749	0.0759	0.0753	0.914
	β_1	0.0985	0.0974	0.0970	0.0981	0.0975	0.955
	β_2	0.1146	0.1197	0.1193	0.1209	0.1199	0.968
Binomial ($n = 2000$)	α_1	0.0484	0.0480	0.0479	0.0481	0.0480	0.943
	α_2	0.0494	0.0478	0.0478	0.0480	0.0479	0.942
	α_3	0.0484	0.0479	0.0479	0.0481	0.0480	0.945
	β_1	0.0609	0.0612	0.0611	0.0614	0.0613	0.959
	β_2	0.0748	0.0746	0.0745	0.0748	0.0746	0.942

Empirical standard deviation (emp.sd), model-based standard error (se.mod), robust sandwich standard error (se.rob), leverage-adjusted standard error (se.lev), finite-sample corrected standard error (se.fin), and coverage probability of the 95% confidence interval based on the model-based variance estimator (cov.prob.mod) for (α, β) . Results are based on 1000 Monte Carlo samples with sample sizes $n = 400$, $n = 800$, and $n = 2000$.